

Zhihao Zhu

Postdoctoral Fellow at HKUST

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Experience

- **Postdoctoral Fellow**, Hong Kong University of Science and Technology, advised by Prof. Yi Yang.
- **Ph.D. in Computer Science**, University of Science and Technology of China (2025), advised by Prof. Defu Lian.
- **Bachelor's degree in Computer Science**, Fuzhou University (2020).

Research Focus

My research centers on trustworthy AI and large language models, with a focus on auditing, privacy, recommender systems, and training data exposure. I develop methods to audit user data revocation, membership inference, and model stealing risks in recommender and graph learning systems. I also study how modern language and vision-language models expose pre-training data, and build frameworks for detecting such leakage via neuron activations and parameter gradients.

A third theme of my work is robust learning: stress-testing graph classification and recommendation services against adversarial attacks, ownership verification, and attribute unlearning, with the goal of making deployed ML systems more accountable and secure.

Trustworthy AI

Auditing model behavior, data revocation, membership inference, and model stealing risks in recommender and graph learning systems.

Large Language Models

Detecting training data exposure and understanding privacy risks in modern language and vision-language models.

Business & Interdisciplinary Journals

Forget Me If You Can: Auditing User Data Revocation in Recommendation Systems

Zhihao Zhu, Yi Yang, Yangyang Fan, Defu Lian

Information Systems Research 2026

HoneyImage: Verifiable, Harmless, and Stealthy Dataset Ownership Verification for Image Models

Zhihao Zhu, Jiale Han, Yi Yang

Management Information Systems Quarterly Major Revision

Revealing Training Data Exposure in Vision-Language Large Models via Parameter Gradients

Zhihao Zhu, Hongyi Tang, Yi Yang, Ahmed Abbasi

Nature Communications Major Revision

RecShield: Output-Level Attribute Unlearning in Recommender Systems

Zhihao Zhu, Yi Yang

Information Systems Research Under Review

GraphMSA: Stress Testing Graph Classification Services Against Model Stealing Attacks

Zhihao Zhu, Yi Yang, Chenwang Wu, Defu Lian

INFORMS Journal on Computing Under Review

AI Conferences and Journals

TDDBench: A Benchmark for Training Data Detection

Zhihao Zhu, Yi Yang, Defu Lian

International Conference on Learning Representations 2025

Identifying Pre-training Data in LLMs: A Neuron Activation-Based Detection Framework

Hongyi Tang*, Zhihao Zhu*, Yi Yang. Equal contribution.

Conference on Empirical Methods in Natural Language Processing 2025

Membership Inference Attacks against Sequential Recommender Systems

Zhihao Zhu, Chenwang Wu, Rui Fan, Defu Lian, Enhong Chen

The Web Conference 2023